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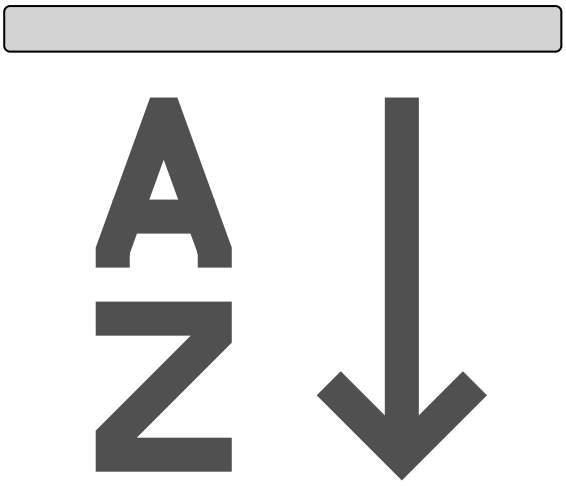

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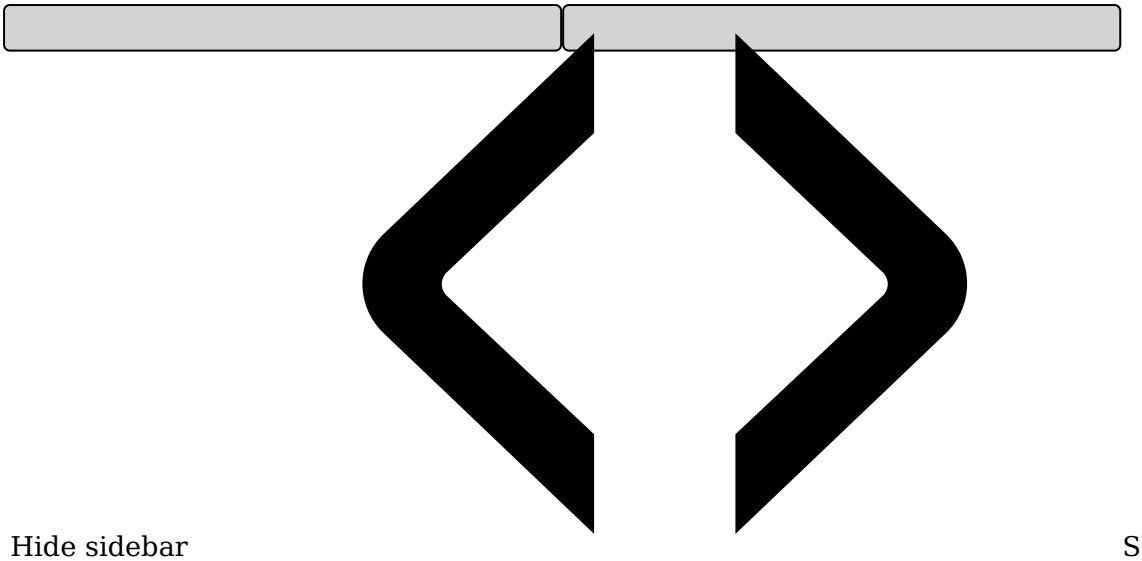

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Enum Class Thread.State

java.lang.Object
java.lang.Enum<Thread.State>

java.lang.Thread.State

All Implemented Interfaces:
Serializable, Comparable<Thread.State>, Constable

Enclosing class:
Thread

---

public static enum Thread.State extends Enum<Thread.State>
A thread state. A thread can be in one of the following states:

- NEW
  A thread that has not yet started is in this state.
- RUNNABLE
  A thread executing in the Java virtual machine is in this state.
- BLOCKED
  A thread that is blocked waiting for a monitor lock is in this state.
- WAITING
  A thread that is waiting indefinitely for another thread to perform a particular action is in this state.
- TIMED_WAITING
  A thread that is waiting for another thread to perform an action for up to a specified waiting time is in this state.
- TERMINATED
  A thread that has exited is in this state.

A thread can be in only one state at a given point in time. These states are virtual machine states which do not reflect any operating system thread states.

Since:
1.5
See Also:
- Thread.getState()

Nested Class Summary

Nested classes/interfaces declared in class Enum

Enum.EnumDesc<E>
Modifier and Type
Class
Description
static final class
Enum.EnumDesc<E extends Enum<E>>
A nominal descriptor for an enum constant.

Enum Constant Summary

Enum Constants
Enum Constant

Description
BLOCKED
Thread state for a thread blocked waiting for a monitor lock.
NEW
Thread state for a thread which has not yet started.
RUNNABLE
Thread state for a runnable thread.
TERMINATED
Thread state for a terminated thread.
TIMED_WAITING
Thread state for a waiting thread with a specified waiting time.
WAITING
Thread state for a waiting thread.

Method Summary

All Methods | Static Methods | Concrete Methods

Modifier and Type
Method
Description
`static Thread.State`
`valueOf(String name)`
Returns the enum constant of this class with the specified name.
`static Thread.State[]`
`values()`
Returns an array containing the constants of this enum class, in the order they are declared.

Methods declared in class Enum

clone, compareTo, describeConstable, equals, finalize, getDeclaringClass, hashCode, name, ordinal, toString, valueOf

Modifier and Type
Method
Description
`protected final Object`
`clone()`
Throws CloneNotSupportedException.
`final int`
`compareTo(Thread.State o)`
Compares this enum with the specified object for order.
`final Optional<Enum.EnumDesc<Thread.State>>`
`describeConstable()`
Returns an enum descriptor `EnumDesc` for this instance, if one can be constructed, or an empty `Optional` if one cannot be.
`final boolean`
`equals(Object other)`
Returns true if the specified object is equal to this enum constant.
`protected final void`

finalize()
Deprecated, for removal: This API element is subject to removal in a future version.
Finalization has been deprecated for removal.
final Class<Thread.State>
getDeclaringClass()
Returns the Class object corresponding to this enum constant's enum type.
final int
hashCode()
Returns a hash code for this enum constant.
final String
name()
Returns the name of this enum constant, exactly as declared in its enum declaration.
final int
ordinal()
Returns the ordinal of this enumeration constant (its position in its enum declaration, where the initial constant is assigned an ordinal of zero).
String
toString()
Returns the name of this enum constant, as contained in the declaration.
static <T extends Enum<T>>
T
valueOf(Class<T> enumClass, String name)
Returns the enum constant of the specified enum class with the specified name.

Methods declared in class Object

getClass, notify, notifyAll, wait, wait, wait
Modifier and Type
Method
Description
final Class<?>
getClass()
Returns the runtime class of this Object.
final void
notify()
Wakes up a single thread that is waiting on this object's monitor.
final void
notifyAll()
Wakes up all threads that are waiting on this object's monitor.
final void
wait()
Causes the current thread to wait until it is awakened, typically by being *notified* or *interrupted*.
final void
wait(long timeoutMillis)

Causes the current thread to wait until it is awakened, typically by being *notified* or *interrupted*, or until a certain amount of real time has elapsed.

`final void`
`wait(long timeoutMillis, int nanos)`
Causes the current thread to wait until it is awakened, typically by being *notified* or *interrupted*, or until a certain amount of real time has elapsed.

Enum Constant Details

NEW

public static final Thread.State NEW
Thread state for a thread which has not yet started.

RUNNABLE

public static final Thread.State RUNNABLE
Thread state for a runnable thread. A thread in the runnable state is executing in the Java virtual machine but it may be waiting for other resources from the operating system such as processor.

BLOCKED

public static final Thread.State BLOCKED
Thread state for a thread blocked waiting for a monitor lock. A thread in the blocked state is waiting for a monitor lock to enter a synchronized block/method or reenter a synchronized block/method after calling `Object.wait`.

WAITING

public static final Thread.State WAITING
Thread state for a waiting thread. A thread is in the waiting state due to calling one of the following methods:

- `Object.wait` with no timeout
- `Thread.join` with no timeout
- `LockSupport.park`

A thread in the waiting state is waiting for another thread to perform a particular action. For example, a thread that has called `Object.wait()` on an object is waiting for another thread to call `Object.notify()` or `Object.notifyAll()` on that object. A thread that has called `Thread.join()` is waiting for a specified thread to terminate.

TIMED_WAITING

public static final Thread.State TIMED_WAITING

Thread state for a waiting thread with a specified waiting time. A thread is in the timed waiting state due to calling one of the following methods with a specified positive waiting time:

- `Thread.sleep`
- `Object.wait` with timeout
- `Thread.join` with timeout
- `LockSupport.parkNanos`
- `LockSupport.parkUntil`

TERMINATED

public static final Thread.State TERMINATED
Thread state for a terminated thread. The thread has completed execution.

Method Details

values

public static Thread.State[] values()
Returns an array containing the constants of this enum class, in the order they are declared.
Returns:
an array containing the constants of this enum class, in the order they are declared

valueOf

public static Thread.State valueOf(String name)
Returns the enum constant of this class with the specified name. The string must match *exactly* an identifier used to declare an enum constant in this class. (Extraneous whitespace characters are not permitted.)
Parameters:
`name` - the name of the enum constant to be returned.
Returns:
the enum constant with the specified name
Throws:
`IllegalArgumentException` - if this enum class has no constant with the specified name
`NullPointerException` - if the argument is null

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual
overviews, definitions of terms, workarounds, and working code examples. Other versions.
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